

-- lab 10 pl/sql tasks

set serveroutput on;

-- task 1

declare

 v_emp_id employees.employee_id%type := &enter_emp_id;

 v_sal employees.salary%type;

 v_dept_name departments.department_name%type;

 v_sal_cat varchar2(20);

 v_dept_class varchar2(20);

begin

 select e.salary, d.department_name

 into v_sal, v_dept_name

 from employees e

 join departments d on e.department_id = d.department_id

 where e.employee_id = v_emp_id;

 if v_sal < 5000 then v_sal_cat := 'low';

 elsif v_sal <= 10000 then v_sal_cat := 'medium';

 else v_sal_cat := 'high'; end if;

 if v_dept_name in ('it', 'engineering') then v_dept_class := 'technical';

 else v_dept_class := 'non-technical'; end if;

 dbms_output.put_line('employee id: ' || v_emp_id);

 dbms_output.put_line('salary category: ' || v_sal_cat);

 dbms_output.put_line('department classification: ' || v_dept_class);

end;

/

-- task 2

begin

 for rec in (select employee_id, salary, job_id from employees) loop

 if rec.job_id like 'it%' and rec.salary > 8000 then

 dbms_output.put_line('emp id: ' || rec.employee_id || ' -> technical recognition');

 else

 dbms_output.put_line('emp id: ' || rec.employee_id || ' -> standard classification');

 end if;

 end loop;

end;

/

-- task 3

declare

```

v_sal_level varchar2(20);
v_perf_label varchar2(20);
begin
  for rec in (select employee_id, salary, job_id from employees) loop
    if rec.salary < 5000 then v_sal_level := 'low';
    elsif rec.salary < 10000 then v_sal_level := 'medium';
    else v_sal_level := 'high'; end if;

    if rec.job_id in ('ad_pres', 'it_prog') then v_perf_label := 'good';
    else v_perf_label := 'normal'; end if;

    dbms_output.put_line('emp: ' || rec.employee_id || ' | salary: ' || v_sal_level || ' |
performance: ' || v_perf_label);
  end loop;
end;
/

```

```

-- task 4
declare
  v_status varchar2(20);
begin
  for rec in (select employee_id, salary, commission_pct from employees) loop
    if rec.salary < 5000 then
      if nvl(rec.commission_pct, 0) < 0.1 then v_status := 'high';
      else v_status := 'medium'; end if;
    else
      v_status := 'low';
    end if;
    dbms_output.put_line('emp: ' || rec.employee_id || ' | performance status: ' || v_status);
  end loop;
end;
/

```

```

-- task 5
declare
  v_dep_class varchar2(20);
begin
  for rec in (select department_id, sum(salary) as total, avg(salary) as average from employees
group by department_id) loop
    if rec.total > 500000 then
      if rec.average > 5000 then v_dep_class := 'high';
      else v_dep_class := 'medium'; end if;
    else
      v_dep_class := 'low';
    end if;
  end loop;
end;

```

```

        end if;
        dbms_output.put_line('dept: ' || rec.department_id || ' | budget class: ' || v_dep_class);
    end loop;
end;
/

-- task 6
declare
    v_eligibility varchar2(10);
    v_job_class varchar2(10);
    v_promo varchar2(10);
    v_dept_id number := &enter_dept_id;
begin
    for rec in (select employee_id, salary, job_id from employees where department_id =
v_dept_id) loop
        if rec.salary < 5000 then v_eligibility := 'yes'; else v_eligibility := 'no'; end if;
        if rec.job_id = 'ad_pres' then v_job_class := 'high'; else v_job_class := 'low'; end if;
        if v_job_class = 'low' and v_eligibility = 'yes' then v_promo := 'yes'; else v_promo := 'no';
    end if;
        dbms_output.put_line('emp: ' || rec.employee_id || ' | eligibility: ' || v_eligibility || ' |
promotion: ' || v_promo);
    end loop;
end;
/

```

```

-- task 7
declare
    v_sal_grade varchar2(10);
    v_job_class varchar2(10);
    v_exp_class varchar2(10);
begin
    for rec in (select employee_id, salary, job_id, hire_date from employees) loop
        if rec.salary < 5000 then v_sal_grade := 'low';
        elsif rec.salary < 10000 then v_sal_grade := 'medium';
        else v_sal_grade := 'high'; end if;

        if rec.job_id like '%man%' then v_job_class := 'high'; else v_job_class := 'low'; end if;

        if months_between(sysdate, rec.hire_date) < 60 then v_exp_class := 'junior';
        elsif months_between(sysdate, rec.hire_date) < 120 then v_exp_class := 'mid';
        else v_exp_class := 'senior'; end if;

        dbms_output.put_line('emp: ' || rec.employee_id || ' | salary: ' || v_sal_grade || ' | job: ' ||
v_job_class || ' | experience: ' || v_exp_class);
    end loop;
end;
/

```

```
    end loop;
end;
/
```

```
-- task 8
```

```
declare
```

```
    v_bonus varchar2(10);
```

```
begin
```

```
    for rec in (select employee_id, salary, commission_pct from employees) loop
```

```
        if rec.salary < 5000 and nvl(rec.commission_pct, 0) > 0.1 then v_bonus := 'yes';
```

```
        else v_bonus := 'no'; end if;
```

```
        dbms_output.put_line('emp: ' || rec.employee_id || ' | bonus eligible: ' || v_bonus);
```

```
    end loop;
```

```
end;
```

```
/
```

```
-- task 9
```

```
declare
```

```
    v_avg_sal number;
```

```
    v_job_class varchar2(10);
```

```
begin
```

```
    select avg(salary) into v_avg_sal from employees;
```

```
    for rec in (select employee_id, salary, job_id from employees) loop
```

```
        if rec.salary < v_avg_sal then
```

```
            if rec.job_id like '%man%' then v_job_class := 'high'; else v_job_class := 'low'; end if;
```

```
            dbms_output.put_line('emp: ' || rec.employee_id || ' | job class: ' || v_job_class || ' | status:
```

```
below average');
```

```
        end if;
```

```
    end loop;
```

```
end;
```

```
/
```

```
-- task 10
```

```
declare
```

```
    v_tier varchar2(10);
```

```
    v_cat varchar2(10);
```

```
    v_dep_class varchar2(10);
```

```
begin
```

```
    for rec in (select employee_id, salary, job_id, department_id from employees order by salary desc) loop
```

```
        if rec.salary < 5000 then v_tier := 'low';
```

```
        elsif rec.salary < 10000 then v_tier := 'medium';
```

```
        else v_tier := 'high'; end if;
```

```

        if rec.job_id like '%man%' then v_cat := 'high'; else v_cat := 'low'; end if;

        if rec.department_id < 50 then v_dep_class := 'a'; else v_dep_class := 'b'; end if;

        dbms_output.put_line('emp: ' || rec.employee_id || ' | tier: ' || v_tier || ' | job cat: ' || v_cat || ' |
dept: ' || v_dep_class);
    end loop;
end;
/

```

-- task 11

```

declare
    v_sal_status varchar2(10);
    v_job_class varchar2(10);
begin
    for dept in (select distinct department_id from employees) loop
        dbms_output.put_line('=== department: ' || dept.department_id || ' ===');
        for emp in (select employee_id, salary, job_id from employees where department_id =
dept.department_id) loop
            if emp.salary < 5000 then v_sal_status := 'low'; else v_sal_status := 'high'; end if;
            if emp.job_id like '%man%' then v_job_class := 'high'; else v_job_class := 'low'; end if;
            dbms_output.put_line('emp: ' || emp.employee_id || ' | salary: ' || v_sal_status || ' | job: ' ||
v_job_class);
        end loop;
    end loop;
end;
/

```

-- task 12

```

declare
    v_emp_id number := &enter_emp_id;
    v_sal number;
    v_job varchar2(20);
    v_grade varchar2(10);
    v_class varchar2(10);
    v_promo varchar2(10);
begin
    select salary, job_id into v_sal, v_job from employees where employee_id = v_emp_id;

    if v_sal < 5000 then v_grade := 'low'; else v_grade := 'high'; end if;
    if v_job like '%man%' then v_class := 'high'; else v_class := 'low'; end if;
    if v_sal < 5000 then v_promo := 'yes'; else v_promo := 'no'; end if;

```

```

        dbms_output.put_line('emp: ' || v_emp_id || ' | grade: ' || v_grade || ' | class: ' || v_class || ' |
promo: ' || v_promo);
end;
/

```

-- task 13

declare

 v_inc number := 0.05;

begin

 for rec in (select employee_id, department_id, hire_date from employees) loop

 if rec.department_id = 10 then v_inc := 0.10;

 elsif months_between(sysdate, rec.hire_date) > 120 then v_inc := 0.08;

 else v_inc := 0.05; end if;

 update employees set salary = salary + (salary * v_inc) where employee_id =
rec.employee_id;

 dbms_output.put_line('updated emp: ' || rec.employee_id || ' with increment: ' || (v_inc*100)
|| '%');

 end loop;

 rollback; -- remove this line in production to commit changes

end;

/

-- task 14

declare

 v_exp varchar2(10);

 v_grade varchar2(10);

begin

 for rec in (select employee_id, hire_date, salary, job_id from employees) loop

 if months_between(sysdate, rec.hire_date) < 60 then v_exp := 'junior';

 elsif months_between(sysdate, rec.hire_date) < 120 then v_exp := 'mid';

 else v_exp := 'senior'; end if;

 if rec.salary < 5000 then v_grade := 'low'; else v_grade := 'high'; end if;

 dbms_output.put_line('emp: ' || rec.employee_id || ' | experience: ' || v_exp || ' | grade: ' ||
v_grade || ' | role: ' || rec.job_id);

 end loop;

end;

/

-- task 15

declare

 v_grade varchar2(10);

```

v_class varchar2(10);
v_bonus varchar2(10);
begin
  for rec in (select employee_id, salary, job_id, commission_pct from employees) loop
    if rec.salary < 5000 then v_grade := 'low'; else v_grade := 'high'; end if;
    if rec.job_id like '%man%' then v_class := 'high'; else v_class := 'low'; end if;
    if nvl(rec.commission_pct, 0) > 0.1 then v_bonus := 'yes'; else v_bonus := 'no'; end if;

    dbms_output.put_line('emp: ' || rec.employee_id || ' | grade: ' || v_grade || ' | class: ' ||
v_class || ' | bonus: ' || v_bonus);
  end loop;
end;
/

```

-- task 16

create or replace procedure show_emp_details(p_emp_id number) is

```

  v_fname employees.first_name%type;
  v_jid employees.job_id%type;
  v_sal employees.salary%type;
  v_did employees.department_id%type;
  v_hdate employees.hire_date%type;
begin
  select first_name, job_id, salary, department_id, hire_date
  into v_fname, v_jid, v_sal, v_did, v_hdate
  from employees where employee_id = p_emp_id;

  dbms_output.put_line('emp id: ' || p_emp_id);
  dbms_output.put_line('first name: ' || v_fname);
  dbms_output.put_line('job id: ' || v_jid);
  dbms_output.put_line('salary: ' || v_sal);
  dbms_output.put_line('dept id: ' || v_did);
  dbms_output.put_line('hire date: ' || v_hdate);
end;
/

```

-- task 17

create or replace procedure increase_salary(p_emp_id number, p_pct number) is

```

  v_new_sal number;
  v_name varchar2(30);
begin
  update employees set salary = salary + (salary * (p_pct / 100)) where employee_id =
p_emp_id;
  select first_name, salary into v_name, v_new_sal from employees where employee_id =
p_emp_id;

```

```

    dbms_output.put_line('emp: ' || v_name || ' | new salary: ' || v_new_sal);
end;
/

```

-- task 18

```

create or replace procedure list_dept_emps(p_dept_id number) is
begin
    dbms_output.put_line('employees in dept: ' || p_dept_id);
    for rec in (select employee_id, job_id, salary from employees where department_id =
p_dept_id) loop
        dbms_output.put_line('id: ' || rec.employee_id || ' | job: ' || rec.job_id || ' | salary: ' ||
rec.salary);
    end loop;
end;
/

```

-- task 19

```

create or replace procedure dept_salary_stats(p_dept_id number) is
    v_count number;
    v_total number;
    v_avg number;
begin
    select count(*), sum(salary), avg(salary) into v_count, v_total, v_avg
    from employees where department_id = p_dept_id;

    dbms_output.put_line('dept: ' || p_dept_id);
    dbms_output.put_line('total emps: ' || v_count);
    dbms_output.put_line('total salary: ' || v_total);
    dbms_output.put_line('avg salary: ' || v_avg);
end;
/

```

-- task 20

```

create or replace procedure list_by_job(p_job_id varchar2) is
begin
    dbms_output.put_line('employees with job: ' || p_job_id);
    for rec in (
        select e.employee_id, e.salary, e.hire_date, d.department_name
        from employees e join departments d on e.department_id = d.department_id
        where e.job_id = p_job_id
    ) loop
        dbms_output.put_line('emp: ' || rec.employee_id || ' | dept: ' || rec.department_name || ' |
salary: ' || rec.salary || ' | hired: ' || rec.hire_date);
    end loop;
end;

```

```
end;  
/
```